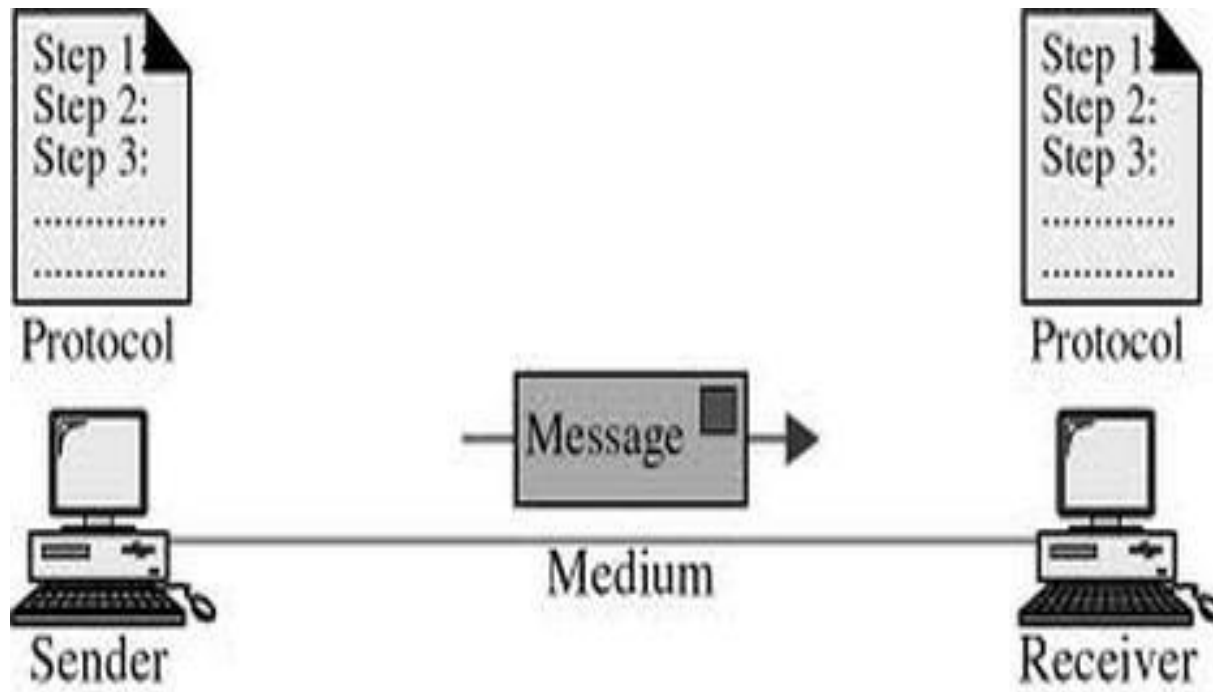


Data Communication

- Data means information in digital format and communication means to exchange information between two or many users in any ways like speaking, texting, or any other modes of the medium. So, data communication is simply the exchange of data between two or many users through the transmission media like twisted pair cable, coaxial cable, optical fibers, radio wave, satellite microwave, etc. The user or device that sends the data is the source and the user that receives the data is a receiver. For data interchange to take place, the communicating devices must be a part of a system comprising of a combination of hardware and software. Its efficiency depends on four fundamental characteristics.
- **The effectiveness depends on four fundamental characteristics of data communications**
- **1. Delivery:** The system must deliver data to the correct destination.
2. Accuracy: The system must deliver data without error and accurately.
3. Timeliness: The system must deliver data in a timely manner. Late delivered data is useless.
4. Jitter: It is the variation in the packet arrival time.

Components of data communication system

- A Communication system has following components:
- 1. **Message**: It is the information or data to be communicated. It can consist of text, numbers, pictures, sound or video or any combination of these.
- 2. **Sender**: It is the device/[computer](#) that generates and sends that message.
- 3. **Receiver**: It is the device or [computer](#) that receives the message. The location of receiver computer is generally different from the sender computer. The distance between sender and receiver depends upon the types of network used in between.
- 4. **Medium**: It is the channel or physical path through which the message is carried from sender to the receiver. The medium can be wired like twisted pair wire, coaxial cable, fiber-optic cable or wireless like laser, radio waves, and microwaves.
- 5. **Protocol**: It is a set of rules that govern the communication between the devices. Both sender and receiver follow same protocols to communicate with each other. In Order to make communication successful between devices , some rules and procedures should be agreed upon at the sending and receiving ends of the system. Such rules and procedures are called as Protocols . Different types of protocols are used for different types of communication.



Data Communication system components

A protocol performs the following functions:

1. **Data sequencing.** It refers to breaking a long message into smaller packets of fixed size. Data sequencing rules define the method of numbering packets to detect loss or duplication of packets, and to correctly identify packets, which belong to same message.
- 2. **Data routing.** Data routing defines the most efficient path between the source and destination.
- 3. **Data formatting.** Data formatting rules define which group of bits or characters within packet constitute data, control, addressing, or other information.
- 4. **Flow control.** A communication protocol also prevents a fast sender from overwhelming a slow receiver. It ensures resource sharing and protection against traffic congestion by regulating the flow of data on communication lines.
- 5. **Error control.** These rules are designed to detect errors in messages and to ensure transmission of correct messages. The most common method is to retransmit erroneous message block. In such a case, a block having error is discarded by the receiver and is retransmitted by the sender.

- **6. Precedence and order of transmission.** These rules ensure that all the nodes get a chance to use the communication lines and other resources of the network based on the priorities assigned to them.
- **7. Connection establishment and termination.** These rules define how connections are established, maintained and terminated when two nodes of a network want to communicate with each other.
- **8. Data security.** Providing data security and privacy is also built into most communication software packages. It prevents access of data by unauthorized users.
- **9. Log information.** Several communication software are designed to develop log information, which consists of all jobs and data communications tasks that have taken place. Such information may be used for charging the users of the network based on their usage of the network resources

Transmission modes

- The way in which data is transmitted from one device to another device is known as **transmission mode**.
- The transmission mode is also known as the communication mode.
- Each communication channel has a direction associated with it, and transmission media provide the direction. Therefore, the transmission mode is also known as a directional mode.
- The transmission mode is defined in the physical layer.
- The Transmission mode is divided into three categories:
 - Simplex mode
 - Half-duplex mode
 - Full-duplex mode

Simplex mode

- In Simplex mode, the communication is unidirectional, i.e., the data flow in one direction.
- A device can only send the data but cannot receive it or it can receive the data but cannot send the data.
- This transmission mode is not very popular as mainly communications require the two-way exchange of data. The simplex mode is used in the business field as in sales that do not require any corresponding reply.
- The radio station is a simplex channel as it transmits the signal to the listeners but never allows them to transmit back.
- Keyboard and Monitor are the examples of the simplex mode as a keyboard can only accept the data from the user and monitor can only be used to display the data on the screen.
- The main advantage of the simplex mode is that the full capacity of the communication channel can be utilized during transmission.
- Advantage of Simplex mode:
- In simplex mode, the station can utilize the entire bandwidth of the communication channel, so that more data can be transmitted at a time. Bandwidth: **The maximum amount of data transmitted over an internet connection in a given amount of time.** Speed refers to the maximum rate you can transmit data, typically measured as megabits per second (Mbps). Bandwidth refers to the maximum amount of data your connection can handle at any moment, also measured as Mbps (and increasingly Gbps, for gigabyte connections). Example jar, funnel, water.
- **throughput is an actual measure of how much data is successfully transferred from source to destination**, and bandwidth is a theoretical measure of how much data could be transferred from source to destination. **Throughput measures speed while bandwidth is only indirectly related to speed.** Example pressure cooker. **bandwidth** refers to maximum **throughput**.
- Disadvantage of Simplex mode:
- Communication is unidirectional, so it has no inter-communication between devices.
-

Half-Duplex mode

- In a Half-duplex channel, direction can be reversed, i.e., the station can transmit and receive the data as well.
- Messages flow in both the directions, but not at the same time.
- The entire bandwidth of the communication channel is utilized in one direction at a time.
- In half-duplex mode, it is possible to perform the error detection, and if any error occurs, then the receiver requests the sender to retransmit the data.
- A **Walkie-talkie** is an example of the Half-duplex mode. In Walkie-talkie, one party speaks, and another party listens. After a pause, the other speaks and first party listens. Speaking simultaneously will create the distorted sound which cannot be understood.
- Advantage of Half-duplex mode:
- In half-duplex mode, both the devices can send and receive the data and also can utilize the entire bandwidth of the communication channel during the transmission of data.
- Disadvantage of Half-Duplex mode:
- In half-duplex mode, when one device is sending the data, then another has to wait, this causes the delay in sending the data at the right time.

Full-duplex mode

- In Full duplex mode, the communication is bi-directional, i.e., the data flow in both the directions.
- Both the stations can send and receive the message simultaneously.
- Full-duplex mode has two simplex channels. One channel has traffic moving in one direction, and another channel has traffic flowing in the opposite direction.
- The Full-duplex mode is the fastest mode of communication between devices.
- The most common example of the full-duplex mode is a telephone network. When two people are communicating with each other by a telephone line, both can talk and listen at the same time.
- Advantage of Full-duplex mode:
 - Both the stations can send and receive the data at the same time.
- Disadvantage of Full-duplex mode:
 - No proper bandwidth utilization as the same line is used for sending and receiving data at the same time.

Differences b/w Simplex, Half-duplex and Full-duplex mode

asis for comparison	Simplex mode	Half-duplex mode	Full-duplex mode
Direction of communication	In simplex mode, the communication is unidirectional.	In half-duplex mode, the communication is bidirectional, but one at a time.	In full-duplex mode, the communication is bidirectional.
Send/Receive	A device can only send the data but cannot receive it or it can only receive the data but cannot send it.	Both the devices can send and receive the data, but one at a time.	Both the devices can send and receive the data simultaneously.
Performance	The performance of half-duplex mode is better than the simplex mode.	The performance of full-duplex mode is better than the half-duplex mode.	The Full-duplex mode has better performance among simplex and half-duplex mode as it doubles the utilization of the capacity of the communication channel.
Example	Examples of Simplex mode are radio, keyboard, and monitor.	Example of half-duplex is Walkie-Talkies.	Example of the Full-duplex mode is a telephone network.

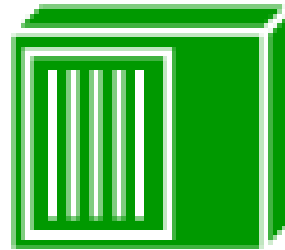
- A computer network is a group of computers linked to each other that enables the computer to communicate with another computer and share their resources, data, and applications.
- A **computer** is a machine that can be programmed to carry out sequences of arithmetic or logical operations automatically. Modern **computers** can perform generic sets of operations known as programs.

Basics of Computer Networking

- **Open system:**
A system which is connected to the network and is ready for communication.
- **Closed system:**
A system which is not connected to the network and can't be communicated with.
- **Computer Network:**
An interconnection of multiple devices, also known as hosts, that are connected using multiple paths for the purpose of sending/receiving data or media. Computer networks can also include multiple devices/mediums which help in the communication between two different devices; these are known as **Network devices** and include things such as routers, switches, hubs, and bridges.
-



Router



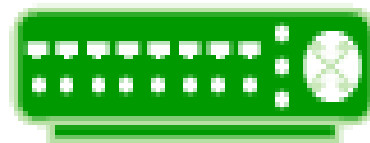
Hub



Bridge



Wireless
Router



Switch



Wireless
Bridge

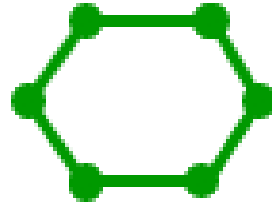
- **Network Topology:**

The layout arrangement of the different devices in a network. Common examples include: Bus, Star, Mesh, Ring, and Daisy chain.

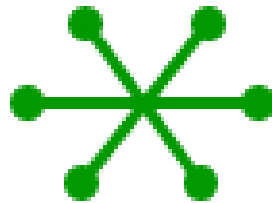
Point-to-Point



Ring



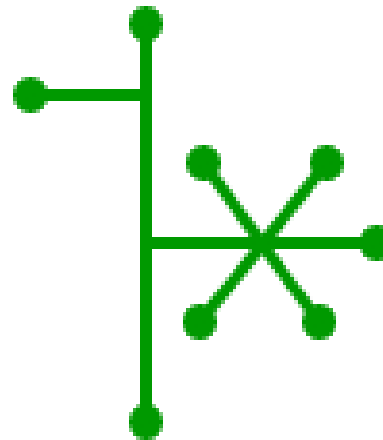
star



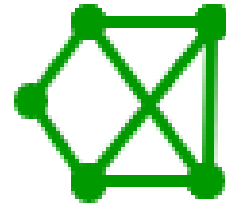
Daisy Chain



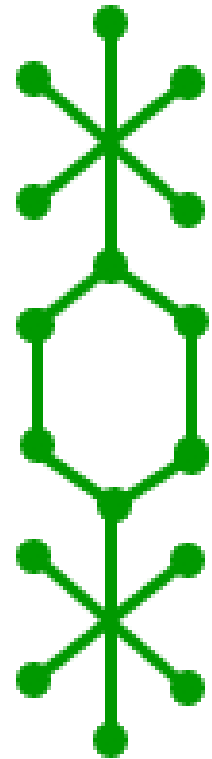
Tree



Mesh



Hybrid



- **OSI:**
OSI stands for **Open Systems Interconnection**. It is a reference model that specifies standards for communications protocols and also the functionalities of each layer.
- **Protocol:**
A protocol is the set of rules or algorithms which define the way how two entities can communicate across the network and there exists different protocol defined at each layer of the OSI model. Few of such protocols are TCP, IP, UDP, ARP, DHCP, FTP and so on.
- **UNIQUE IDENTIFIERS OF NETWORK**
Host name:
Each device in the network is associated with a unique device name known as Hostname.
Type “hostname” in the command prompt(Administrator Mode) and press ‘Enter’, this displays the hostname of your machine.

Administrator: Command Prompt

Microsoft Windows [Version 6.2.9200]
(c) 2012 Microsoft Corporation. All rights reserved.

C:\Windows\system32>hostname
kundana

C:\Windows\system32>

- **IP Address (Internet Protocol address):**

Also known as the Logical Address, the IP Address is the network address of the system across the network.

To identify each device in the world-wide-web, the Internet Assigned Numbers Authority (IANA) assigns an IPV4 (Version 4) address as a unique identifier to each device on the Internet.

The length of an IPv4 address is 32-bits, hence, we have 2^{32} IP addresses available. The length of an IPv6 address is 128-bits.

Type “ipconfig” in the command prompt and press ‘Enter’, this gives us the IP address of the device.

- **MAC Address (Media Access Control address):**

Also known as physical address, the MAC Address is the unique identifier of each host and is associated with its NIC (Network Interface Card). Network Interface Card (NIC) is a **hardware component** that is present on the computer. It is used to **connect different networking devices** such as computers and servers to share data over the connected network.

A MAC address is assigned to the NIC at the time of manufacturing.

The length of the MAC address is : 12-nibble/ 6 bytes/ 48 bits

Type “ipconfig/all” in the command prompt and press ‘Enter’, this gives us the MAC address.

- **Port:**

A port can be referred to as a logical channel through which data can be sent/received to an application. Any host may have multiple applications running, and each of these applications is identified using the port number on which they are running.

A port number is a 16-bit integer, hence, we have 2^{16} ports available which are categorized as shown below:

Port Types

- Range
- Well known Ports 0 – 1023
- Registered Ports 1024 – 49151
- Ephemeral Ports 49152 – 65535
- Number of ports: 65,536

Range: 0 – 65535

Type “netstat -a” in the command prompt and press ‘Enter’, this lists all the ports being used.

Port-443 allows data transmission over a secured network, while **Port 80** enables data transmission in plain text. Users will get an insecure warning if he tries to access a non-HTTPS web page. **Port 443** encrypts network data packets before data transmission takes place.08-Mar-2021

Administrator: Command Prompt

```
C:\Windows\system32>netstat -a
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	kundana:0	LISTENING
TCP	0.0.0.0:443	kundana:0	LISTENING
TCP	0.0.0.0:445	kundana:0	LISTENING
TCP	0.0.0.0:902	kundana:0	LISTENING
TCP	0.0.0.0:912	kundana:0	LISTENING
TCP	0.0.0.0:8092	kundana:0	LISTENING
TCP	0.0.0.0:49152	kundana:0	LISTENING
TCP	0.0.0.0:49153	kundana:0	LISTENING
TCP	0.0.0.0:49154	kundana:0	LISTENING
TCP	0.0.0.0:49155	kundana:0	LISTENING
TCP	0.0.0.0:49158	kundana:0	LISTENING
TCP	0.0.0.0:49160	kundana:0	LISTENING
TCP	127.0.0.1:5354	kundana:0	LISTENING
TCP	127.0.0.1:5939	kundana:0	LISTENING
TCP	127.0.0.1:8307	kundana:0	LISTENING
TCP	127.0.0.1:30000	kundana:0	LISTENING
TCP	192.168.1.99:139	kundana:0	LISTENING

- **Socket:**

The unique combination of IP address and Port number together are termed as Socket.

- **Other related concepts**

DNS Server:

DNS stands for **Domain Name system**.

DNS is basically a server which translates web addresses or URLs (ex: www.google.com) into their corresponding IP addresses. We don't have to remember all the IP addresses of each and every website.

The command '**nslookup**' gives you the IP address of the domain you are looking for. This also provides the information of our DNS Server.

Administrator: Command Prompt

```
C:\Windows\system32>nslookup www.google.com
```

```
Server: UnKnown
```

```
Address: 192.168.43.1
```

```
Non-authoritative answer:
```

```
Name: www.google.com
```

```
Addresses: 2404:6800:4003:c00::6a
```

```
74.125.200.106
```

```
74.125.200.104
```

```
74.125.200.99
```

```
74.125.200.103
```

```
74.125.200.147
```

```
74.125.200.105
```

```
C:\Windows\system32>
```

- **ARP:**

ARP stands for **Address Resolution Protocol**.

It is used to convert an IP address to its corresponding physical address(i.e., MAC Address).

ARP is used by the Data Link Layer to identify the MAC address of the Receiver's machine.

- **RARP:**

RARP stands for **Reverse Address Resolution Protocol**.

As the name suggests, it provides the IP address of the device given a physical address as input. But RARP has become obsolete since the time DHCP has come into the picture.

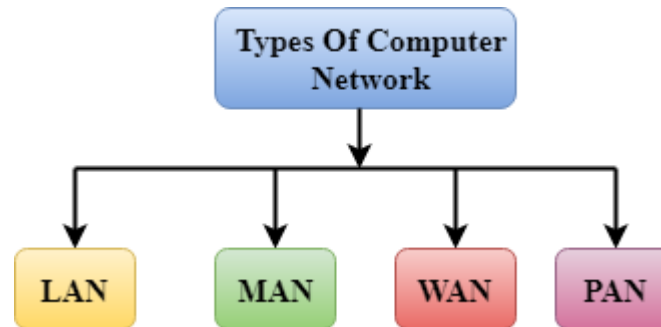
Network types

- LAN(Local Area Network)
- PAN(Personal Area Network)
- MAN(Metropolitan Area Network)
- WAN(Wide Area Network)

A computer network is a group of computers linked to each other that enables the computer to communicate with another computer and share their resources, data, and applications.

A computer network can be categorized by their size. A **computer network** is mainly of **four types**:

- LAN(Local Area Network)
- PAN(Personal Area Network)
- MAN(Metropolitan Area Network)
- WAN(Wide Area Network)



LAN(Local Area Network)

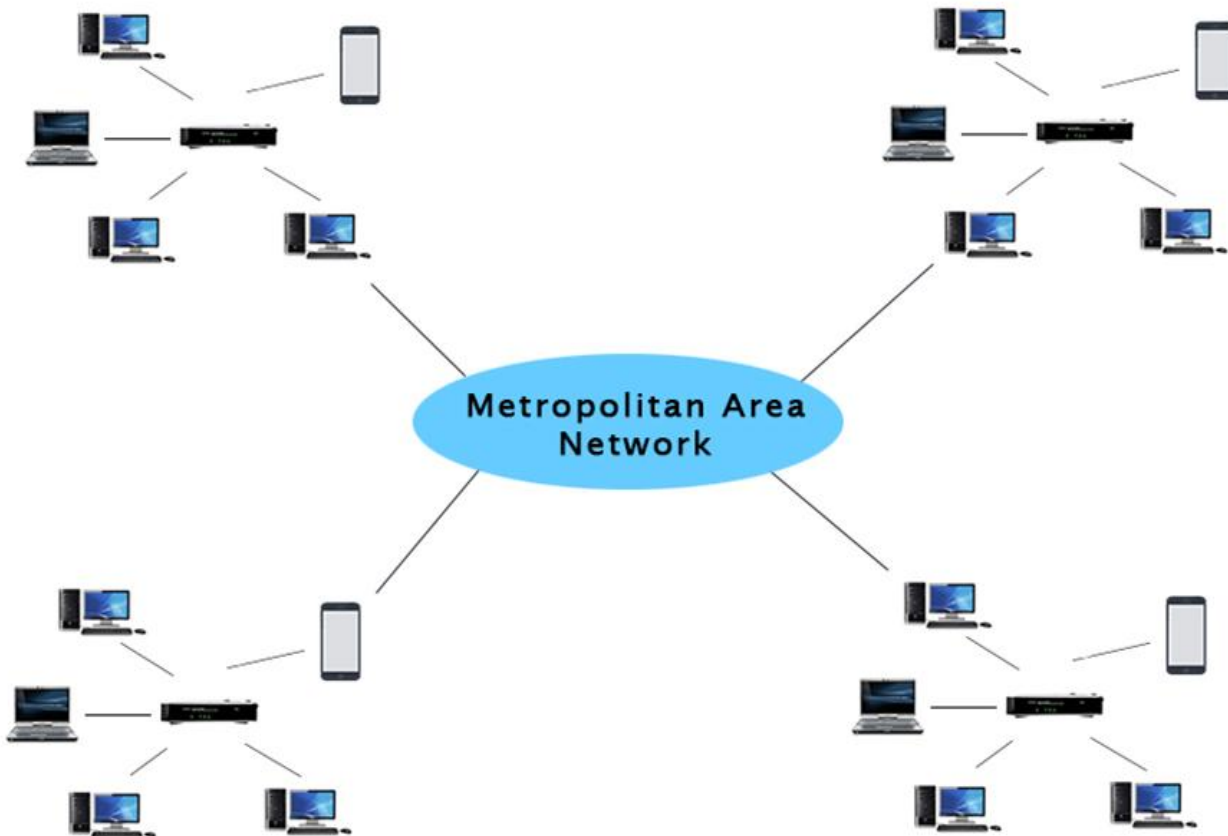
- Local Area Network is a group of computers connected to each other in a small area such as building, office.
- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and ethernet cables.
- The data is transferred at an extremely faster rate in Local Area Network.
- Local Area Network provides higher security.



PAN(Personal Area Network)

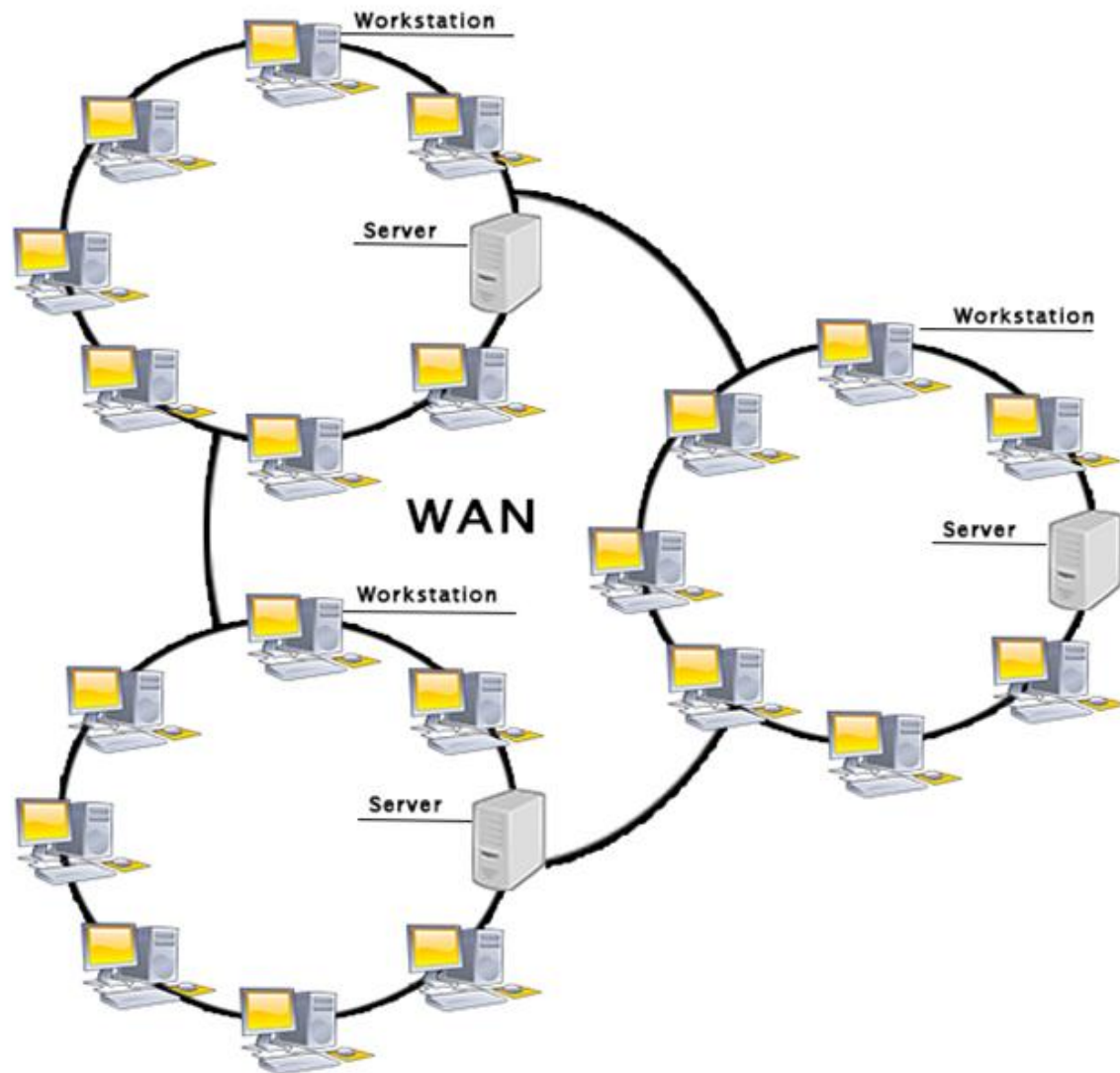
- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- **Thomas Zimmerman** was the first research scientist to bring the idea of the Personal Area Network.
- Personal Area Network covers an area of **30 feet**.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations.





- Uses Of Metropolitan Area Network:
- MAN is used in communication between the banks in a city.
- It can be used in an Airline Reservation.
- It can be used in a college within a city.
- It can also be used for communication in the military.
-

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A Wide Area Network is quite bigger network than the LAN.
- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fibre optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of Business, government, and education.



- Examples Of Wide Area Network:
- **Mobile Broadband:** A 4G network is widely used across a region or country.
- **Last mile:** A telecom company is used to provide the internet services to the customers in hundreds of cities by connecting their home with fiber.
- **Private network:** A bank provides a private network that connects the 44 offices. This network is made by using the telephone leased line provided by the telecom company.
- Advantages Of Wide Area Network:
- Following are the advantages of the Wide Area Network:
- **Geographical area:** A Wide Area Network provides a large geographical area. Suppose if the branch of our office is in a different city then we can connect with them through WAN. The internet provides a leased line through which we can connect with another branch.
- **Centralized data:** In case of WAN network, data is centralized. Therefore, we do not need to buy the emails, files or back up servers.
- **Get updated files:** Software companies work on the live server. Therefore, the programmers get the updated files within seconds.
- **Exchange messages:** In a WAN network, messages are transmitted fast. The web application like Facebook, Whatsapp, Skype allows you to communicate with friends.
- **Sharing of software and resources:** In WAN network, we can share the software and other resources like a hard drive, RAM.
- **Global business:** We can do the business over the internet globally.
- **High bandwidth:** If we use the leased lines for our company then this gives the high bandwidth. The high bandwidth increases the data transfer rate which in turn increases the productivity of our company.
-

- Disadvantages of Wide Area Network:
- The following are the disadvantages of the Wide Area Network:
- **Security issue:** A WAN network has more security issues as compared to LAN and MAN network as all the technologies are combined together that creates the security problem.
- **Needs Firewall & antivirus software:** The data is transferred on the internet which can be changed or hacked by the hackers, so the firewall needs to be used. Some people can inject the virus in our system so antivirus is needed to protect from such a virus.
- **High Setup cost:** An installation cost of the WAN network is high as it involves the purchasing of routers, switches.
- **Troubleshooting problems:** It covers a large area so fixing the problem is difficult.
- Internet is a vast network that connects computers all over the world.